Multiple General Binomial Expansions

Exam code: 9MA0

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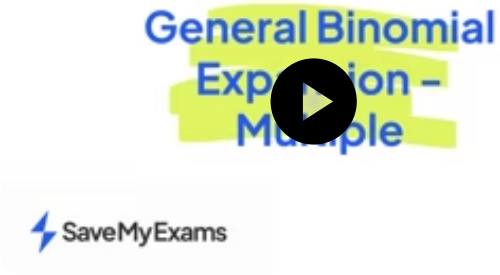
Updated on 20 June 2025



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Multiple general binomial expansions

What is the general binomial expansion?

- The **general binomial expansion**, as given in the formula booklet, is

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1\times 2}x^2 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{1\times 2\times\ldots\times r}x^r + \ldots \qquad (|x|<1,\ n\in\mathbb{R})$$

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- If $n \in \mathbb{N}$ then the expansion is **finite** (see Binomial Expansion)
- Otherwise the expansion is infinitely long
  - It is only valid for $|x| < 1$ ($-1 < x < 1$)
  - Only the first few terms of an expansion are usually needed

What are multiple general binomial expansions?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The expansion will only be valid for the lowest **|x|** boundary from all the expansions used

Help me understand

How do I expand multiple binomials?

STEP 1 Break the expression down into binomial expansions

STEP 2 Expand each binomial individually, up to a suitable number of terms

- Be careful with negatives and fractions
- Use brackets as appropriate

STEP 3 Collect the expansions together and simplify

- This could be expanding brackets, collecting like terms, etc
- Ignore any terms of degree higher than required

STEP 4 Check the validity of each binomial expansion

- The overall validity is the intersection (∩)

e.g. FIND THE BINOMIAL EXPANSION OF $\frac{\sqrt{1+x}}{3+2x}$ UP TO AND INCLUDING THE TERM IN x^3. STATE THE VALUES OF x FOR WHICH THE EXPANSION IS VALID.

$$\frac{\sqrt{1+x}}{3+2x} = (1+x)^{\frac{1}{2}}(3+2x)^{-1}$$

STEP 1: BREAK THE EXPRESSION DOWN INTO BINOMIAL EXPANSIONS

$$(1+x)^{\frac{1}{2}} = 1 + \left(\frac{1}{2}\right)x + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2}x^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{6}x^3 + \ldots$$

USE BRACKETS

$$= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \ldots$$

STEP 2: EXPAND EACH BINOMIAL INDIVIDUALLY

$$(3+2x)^{-1} = 3^{-1}\left(1+\frac{2}{3}x\right)^{-1}$$

$$= \frac{1}{3}\left[1 + (-1)\left(\frac{2}{3}x\right) + \frac{(-1)(-2)}{2}\left(\frac{2}{3}x\right)^2 + \frac{(-1)(-2)(-3)}{6}\left(\frac{2}{3}x\right)^3 + \ldots\right]$$

$$= \frac{1}{3}\left[1 - \frac{2}{3}x + \frac{4}{9}x^2 - \frac{8}{27}x^3 + \ldots\right]$$

$$= \frac{1}{3} - \frac{2}{9}x + \frac{4}{27}x^2 - \frac{8}{81}x^3 + \ldots$$

$$\frac{\sqrt{1+x}}{3+2x} = \left(1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \ldots\right)\left(\frac{1}{3} - \frac{2}{9}x + \frac{4}{27}x^2 - \frac{8}{81}x^3 + \ldots\right)$$

STEP 3: EXPAND BRACKETS IN THE USUAL WAY

$$\approx \frac{1}{3} - \frac{2}{9}x + \frac{4}{27}x^2 - \frac{8}{81}x^3$$
$$+ \frac{1}{6}x - \frac{1}{9}x^2 + \frac{2}{27}x^3$$
$$- \frac{1}{24}x^2 + \frac{1}{36}x^3$$
$$+ \frac{1}{48}x^3$$

x^4 AND HIGHER IGNORED

SEPARATE LINES KEEP IT TIDY

$$= \frac{1}{3} - \frac{1}{18}x - \frac{1}{216}x^2 + \frac{31}{1296}x^3$$

$$\sqrt{1+x} = (1+x)^{\frac{1}{2}} \qquad |x| < 1$$

$$(3+2x)^{-1} = \frac{1}{3}\left(1+\frac{2}{3}x\right)^{-1} \qquad \left|\frac{2}{3}x\right| < 1 \qquad |x| < \frac{3}{2}$$

STEP 4: CHECK THE VALIDITY

$\therefore$ EXPANSION IS ONLY VALID FOR $|x|<1$

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How do I use partial fractions to form multiple general binomial expansions?

- Partial fractions allow rational expressions to be written in a form where the general binomial expansion can then be applied

$$\frac{9x+10}{(x+4)(3x-1)} = \frac{2}{x+4} + \frac{3}{3x-1}$$

WRITE AS PARTIAL FRACTIONS

$$= 2(x+4)^{-1} + 3(3x-1)^{-1}$$

STEP 1: BREAK EXPRESSION DOWN INTO BINOMIAL EXPANSIONS

$$2(x+4)^{-1} = 2(4+x)^{-1}$$
$$= 2[4^{-1}(1+\tfrac{1}{4}x)^{-1}]$$
$$= \tfrac{1}{2}(4+x)^{-1} = \tfrac{1}{2}[1 + (-1)(\tfrac{x}{4}) + \frac{(-1)(-2)}{2}(\tfrac{x}{4})^2 + \ldots]$$
$$= \tfrac{1}{2} - \tfrac{1}{8}x + \tfrac{1}{32}x^2 - \ldots$$

STEP 2: EXPAND EACH BINOMIAL INDIVIDUALLY

$$3(3x-1)^{-1} = 3(-1+3x)^{-1}$$
$$= 3[(-1)^{-1}(1-3x)^{-1}]$$
$$= -3(1-3x)^{-1} = -3[1+(-1)(-3x) + \frac{(-1)(-2)}{2}(-3x)^2 + \ldots]$$
$$= -3 - 9x - 27x^2 - \ldots$$

$$\therefore \frac{9x+10}{(x+4)(3x-1)} = (\tfrac{1}{2} - \tfrac{1}{8}x + \tfrac{1}{32}x^2 - \ldots)$$
$$+(-3 - 9x - 27x^2 - \ldots)$$

STEP 3: COLLECT EXPANSIONS AND SIMPLIFY

LINE UP "LIKE TERMS" MAKES SIMPLIFYING EASIER

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- Validity is an important part of the general binomial expansion

e.g. FOR WHICH VALUES OF x IS THE EXPANSION OF $\frac{9x+10}{(x+4)(3x-1)}$ VALID?

EXPANSIONS USED ARE $(1+\tfrac{1}{4}x)^{-1}$ AND $(1-3x)^{-1}$

STEP 4: CONSIDER VALIDITY OF EXPANSIONS USED SEPARATELY

$(1+\tfrac{1}{4}x)^{-1}$ IS VALID FOR $|\tfrac{1}{4}x| < 1$ $\quad |x| < 4$

$(1-3x)^{-1}$ IS VALID FOR $|-3x| < 1$ $\quad |x| < \tfrac{1}{3}$

$\therefore$ EXPANSION OF $\frac{9x+10}{(x+4)(3x-1)}$ IS VALID FOR $|x| < \tfrac{1}{3}$

STEP 4: THE INTERSECTION ($\cap$) (OVERLAP) OF THE TWO INEQUALITIES

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